AI-Driven Multi-Disease Health Risk Prediction System

Ahmad Baabod
Department of Computer Science
Effat University
Jeddah, Saudi Arabia

Osama Abutaha
Department of Computer Science
Effat University
Jeddah, Saudi Arabia

Albaraa Mashaab
Department of Computer Science
Effat University
Jeddah, Saudi Arabia

Omar Abdulkarim
Department of Computer Science
Effat University
Jeddah, Saudi Arabia

***Abstract*—This paper proposes an AI-driven multi-disease health risk prediction framework that integrates lifestyle and clinical indicators into a unified assessment system. Machine learning models are developed to predict sleep disorders, diabetes, and heart disease using real-world healthcare datasets. Comprehensive preprocessing, class imbalance handling using SMOTE, and rigorous model evaluation are applied. Experimental results demonstrate that ensemble-based models achieve robust predictive performance, highlighting the effectiveness of the proposed system as a decision-support tool for early health risk awareness and preventative healthcare.**

Keywords—Machine learning, healthcare analytics, disease prediction, multi-disease modeling, preventive healthcare

I. Introduction

Chronic non-communicable diseases, including diabetes, sleep disorders, and cardiovascular disease, affect millions worldwide and contribute significantly to morbidity, mortality, and healthcare costs [1][2]. These conditions often develop gradually and may remain undetected until advanced stages, increasing the risk of severe complications. Early identification of high-risk individuals is therefore essential for effective prevention and intervention.

Advances in artificial intelligence (AI) and machine learning (ML) have enabled the analysis of complex healthcare datasets to uncover patterns that traditional statistical methods often fail to detect [3][4]. Recent studies have explored combining genomic and clinical data for disease risk prediction using deep learning frameworks [7], and healthcare predictive analytics surveys show the rapid adoption of ML and DL techniques across multiple disease domains [8].

Previous studies have demonstrated the effectiveness of ML in predicting individual diseases; however, most systems focus on a single condition, limiting their utility for comprehensive health assessment. Explainable multi-disease surveillance systems have recently been developed to combine EHR data and ML models for predicting various chronic conditions simultaneously [9]. Large-scale national datasets have also been used to predict multimorbidity, demonstrating the potential of integrated predictive systems [10].

This paper proposes an AI-driven multi-disease health risk prediction system capable of simultaneously assessing the risk of sleep disorders, diabetes, and heart disease. The system integrates lifestyle attributes, such as sleep duration, stress levels, and physical activity, with clinical indicators including glucose levels, cholesterol, and blood pressure. The primary contributions of this work include:

1. A unified predictive framework for multiple chronic diseases.
2. Robust data preprocessing, including class imbalance handling with SMOTE.
3. Probability-based risk estimation to support early intervention and decision-making.

By combining lifestyle and clinical features, the proposed system enables a more comprehensive assessment, improving early risk detection and preventive healthcare compared to single-disease approaches.

II. Related Work

Machine learning has been widely applied in healthcare for disease prediction and risk assessment.

Diabetes Prediction: Logistic Regression, Random Forest, and Support Vector Machines have been widely used to predict diabetes using clinical biomarkers [3][5]. Pros: High accuracy on clinical datasets. Cons: Often ignore lifestyle factors. Gap: Lack of integration with multi-disease modeling.

Heart Disease Prediction: Ensemble models and probabilistic classifiers have shown strong performance in predicting cardiovascular disease [2][4]. Pros: Captures complex feature interactions. Cons: Most studies predict a single condition, not multiple comorbidities. Gap: Limited integration of lifestyle indicators with clinical data.

Sleep Disorder Prediction: Recent studies highlight correlations between sleep duration, stress, physical activity, and long-term health outcomes [6]. Pros: Focus on lifestyle features. Cons: Small datasets and single-disease focus. Gap: Lack of interpretable and unified multi-disease prediction frameworks.

Recent surveys and reviews highlight the increasing role of machine learning in chronic disease prediction, emphasizing explainable and multi-disease models as a growing research trend [8][11].

In addition, many prior works do not adequately address class imbalance, leading to biased predictions, or provide

interpretable outputs, reducing trust in real-world applications. The proposed system addresses these limitations by developing a multi-disease, explainable, and imbalance-aware prediction framework.

III. Methodology

A. Datasets

Three healthcare datasets were utilized:

- Diabetes: 768 records, 8 clinical features including glucose, BMI, and age.
- Heart Disease: 303 records, 13 attributes including blood pressure, cholesterol, and ECG results.
- Sleep Disorders: 500 records, 10 features combining lifestyle (sleep duration, stress, physical activity) and clinical indicators [10].

B. Data Preprocessing

Data preprocessing involved:

1. Handling missing values using mean imputation [11].
2. Removing duplicates to ensure dataset quality.
3. Encoding categorical variables with one-hot encoding.
4. Normalizing numerical features to a [0,1] range for model compatibility.

To address class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) [6] was applied to the training datasets, improving fairness and predictive performance for minority classes.

C. Model Training

Multiple machine learning models were trained and evaluated:

- Logistic Regression (LR)
- K-Nearest Neighbors (KNN)
- Support Vector Machines (SVM)
- Random Forest (RF)
- Gaussian Naive Bayes (GNB)
- XGBoost

Models were trained using an 80/20 training/testing split. Performance was evaluated using accuracy, precision, recall, F1-score, and probability-based outputs to enable individual risk interpretation.

D. Exploratory Data Analysis (EDA)

Descriptive statistics and visualization were performed to understand feature distributions and detect anomalies. Class distributions revealed imbalance in some datasets, justifying the use of SMOTE.

E. Feature Importance

For ensemble models (RF and XGBoost), feature importance scores were computed to interpret the contribution of each attribute to disease prediction, enhancing transparency and trustworthiness.

IV. Experimental Results

The models were evaluated on the test datasets using accuracy, precision, recall, and F1-score. Ensemble-based models, particularly Random Forest (RF) and XGBoost, achieved superior performance across all three disease prediction tasks.

For diabetes prediction, RF and XGBoost achieved accuracy levels exceeding 85%, with top contributing features including glucose level, BMI, and age. Heart disease prediction was best performed by RF, with cholesterol, blood pressure, and ECG features being most influential. Sleep disorder prediction achieved over 85% accuracy with XGBoost, with sleep duration, stress level, and physical activity being the top features.

Probability-based predictions allowed more detailed interpretation of individual health risk levels, providing a risk score between 0 and 1 for each patient. This supports early intervention and personalized healthcare decision-making.

TABLE I. SUMMARIZES THE PERFORMANCE OF ALL MODELS FOR EACH DISEASE:

Model	**Diabetes Accuracy**	**Heart Disease Accuracy**	**Sleep Disorder Accuracy**
LR	78%	76%	74%
KNN	80%	74%	72%
SVM	81%	77%	75%
RF	87%	85%	84%
GNB	75%	73%	71%
XGBoost	86%	83%	85%

V. Discussion

The results demonstrate that ensemble models outperform traditional classifiers due to their ability to capture non-linear relationships and interactions between features. The use of SMOTE effectively mitigated bias toward majority classes, improving fairness in predictions.

Feature importance analysis highlights the relevance of both clinical and lifestyle attributes in multi-disease prediction. For instance, glucose and BMI are crucial for diabetes prediction, while sleep duration and stress level significantly affect sleep disorder predictions.

Limitations of this study include the relatively small dataset sizes, absence of wearable sensor data, and lack of clinical validation. Despite these limitations, the proposed system provides a strong foundation for a practical decision-

support tool that can guide early interventions and personalized preventive care.

Future improvements may include integrating larger and more diverse datasets, real-time wearable data, and additional explainability methods to enhance transparency and trust.

VI. Conclusion

This paper presents an AI-driven multi-disease health risk prediction system that integrates machine learning and healthcare analytics. The proposed framework effectively predicts sleep disorders, diabetes, and heart disease by combining lifestyle and clinical features, addressing limitations of single-disease prediction systems.

Experimental results indicate that ensemble models, particularly Random Forest and XGBoost, achieve the highest predictive performance. Probability-based risk outputs allow for personalized health risk assessment and early intervention planning.

Future work will focus on incorporating larger datasets, wearable sensor data, and clinical validation to further improve model performance and applicability in real-world healthcare scenarios.

TABLE II. Feature Importance (Top 5 for each disease):

Disease	**Feature 1**	**Feature 2**	**Feature 3**	**Feature 4**	**Feature 5**
Diabetes	Glucose	BMI	Age	Blood Pressure	Cholesterol
Heart Disease	Cholesterol	Blood Pressure	ECG	Age	BMI
Sleep Disorder	Sleep Duration	Stress Level	Physical Activity	Age	BMI

References

[1] I. M. Lee et al., "Effect of physical activity on major non-communicable diseases," The Lancet, 2012.

[2] T. Chandola et al., "Chronic stress and coronary heart disease," BMJ, 2010.

[3] M. Gao et al., "Machine learning approaches for chronic disease prediction," BMC Public Health, 2019.

[4] O. S. Albahri et al., "AI-based healthcare decision support systems: A systematic review," Journal of Biomedical Informatics, 2021.

[5] S. M. Lundberg and S. I. Lee, "A unified approach to interpreting model predictions," NeurIPS, 2017.

[6] N. V. Chawla et al., "SMOTE: Synthetic minority over-sampling technique," JAIR, 2002.

[7] H. Alzoubi, R. Alzubi, and N. Ramzan, "Deep Learning Framework for Complex Disease Risk Prediction Using Genomic Variations.

[8] M. Badawy, N. Ramadan, and H. Hefny, "Healthcare predictive analytics using machine learning and deep learning techniques: a survey," Journal of Electrical Systems and Information Technology.

[9] S. A. Khan et al., "An Explainable Disease Surveillance System for Early Prediction of Multiple Chronic Diseases.

[10] F. M. Albagmi et al., "Predicting Multimorbidity Using Saudi Health Indicators (Sharik) Nationwide Data.

[11] R. Islam and A. Sultana, "A comprehensive review for chronic disease prediction using machine learning algorithms.